

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

Phase 1 18.04.2026				
Team Name: Quanties				
First Name	Last Name	Email	Aqora Username	Role within Team
Grigorii	Vydrevich	grigory.vydrevich@gmail.com	GrigoV	Domain Expert, Coordinator
Victor	Robles	victor.roferz@gmail.com	VictoRoferz	Coder

TEAM EXPERTISE & EXPERIENCE

Building on our previous experience in quantitative finance and quantum simulation, our team combines expertise in **stochastic volatility modeling** and **Noisy Intermediate-Scale Quantum (NISQ) algorithms**.

- **Grigori Vydrevich:** Student at RWTH Aachen in M.Sc. Physics; Experience in Ising dynamics and Quantum Reservoirs with framework from Quandela and QML with PennyLane
- **Victor Robles:** Computer Scientist and Mathematics student at RWTH Aachen University with specialization in AI. Experience working as Quant developer;

PROBLEM STATEMENT & METHODOLOGY

Market volatility is inherently nonstationary, characterized by abrupt regime shifts. Our objective is to design a **Hybrid Quantum Reservoir Computing (QRC)** system that maps these shifts more effectively than classical linear models.

We target a dual-output model: continuous realized-volatility forecasting and discrete regime classification (Calm, Elevated, Stressed). This dual-head approach ensures the model remains grounded in physical market realities rather than abstract quantum states.

TECHNICAL APPROACH

Classical preprocessing pipeline. Relying on our experience in data science, we plan to implement a robust preprocessing stage. Before the quantum reservoir, data undergoes wavelet denoising and PCA-based dimensionality reduction to fit a 5–12 qubit embedding space. Features include log returns, realized volatility, and volume-weighted indicators processed through a strict walk-forward normalization to prevent look-ahead bias.

Quantum architecture. Unlike variational circuits, QRC utilizes the fixed complex dynamics of the quantum system to project inputs into a high-dimensional feature space.

- **Encoding:** Angle-based encoding on input qubits to capture temporal non-linearity.
- **Reservoir:** Fixed Hamiltonian evolution creates a rich “memory” of previous time steps encompassing larger processes.
- **Readout:** Ridge regression classical layer interprets the quantum observables, ensuring the training remains stable and highly interpretable for financial risk management.

BENCHMARKING, SIMULATION, & TIMELINE

A realistic concept requires comparison against established financial industry standards.

- **Baselines:** Comparison against **LSTM** models due to their comparable size and trainability.
- **Simulation Strategy:** Implementation via PennyLane on the qBraid ecosystem, accounting for hardware noise to ensure simulation fidelity matches potential NISQ deployment.

Timeline:

April: Data/Baseline Finalization;

May: QRC Simulator Integration;

June-July: Benchmarking and migration to hardware targets.

REFERENCES

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- Mancilla, J., & Pere, C. (2022). A preprocessing perspective for quantum machine learning classification advantage using NISQ algorithms. arXiv, 2208.13251.
- Otieno, W., Zagoskin, A., Balanov, A. G., Gongora, J. T., & Savel'ev, S. E. (2026). A Quantum Reservoir Computing Approach to Quantum Stock Price Forecasting in Quantum-Invested Markets. arXiv, 2602.13094.